People's Democratic Republic of Algeria

Ministry of Higher Education and Scientific Research

University of Constantine 3

Faculty of Medicine

Physical and chemical properties of Carbohydrates

Course intended for 1st-year medical students

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Course Objectives:

1. List the physical properties of carbohydrates
2. List the chemical properties of carbohydrates
3. Understand the reactivity of carbohydrates according to their hydroxyl and carbonyl functions
4. Use physical and chemical criteria to identify monosaccharides

Course Outline:

I. Introduction

II. Physical properties of monosaccharides

III. Chemical properties of monosaccharides

II.1. Chemical stability of monosaccharides

II.2. Properties related to the carbonyl function

II.3. Properties related to the alcohol function

II.4. Properties related to the proximity of the alcohol and carbonyl functions

Academic Year : 2025–2026

I. INTRODUCTION

Monosaccharides are polyhydroxylated biomolecules that possess a carbonyl group, these two chemical groups confer specific chemical and physical criteria to monosaccharides. Therefore, the knowledge of monosaccharides' behavior in different conditions and their possible reactions with other molecules is important to study monosaccharides.

As a matter of fact, the carbonyl and hydroxyl groups are subject to oxidation. reduction and substitution reactions, which are the basis for synthesis of monosaccharide derivatives.

II. PHYSICAL PROPERTIES

II. 1 Solubility

Monosaccharides are quite soluble in water, each molecule having several OH groups that readily engage in hydrogen bonding with surrounding H_2O molecules.

Inversely, monosaccharides are sparingly soluble in alcohol and insoluble in non-polar organic solvents like ether.

The crystallization of sugars is difficult, in fact, the concentrated aqueous solutions are viscous (syrups). This crystallization can be facilitated by adding alcohol.

II. 2 Optical activity

All monosaccharides, except dihydroxyacetone, have one or more asymmetric carbon atoms and show optical activity.

II. 3 Absorption spectrum

Monosaccharides exhibit poor absorption in the visible and ultraviolet spectrum. In contrast, they possess a characteristic infrared spectrum.

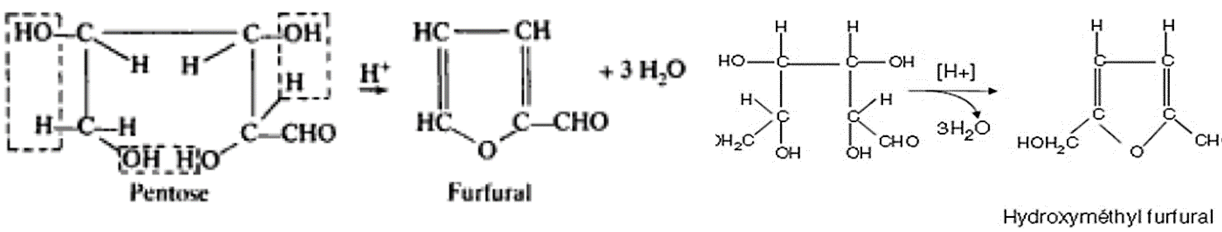
III. CHEMICAL PROPERTIES

II. 1 Chemical stability

a. In acidic conditions

In a diluted acidic environment, monosaccharides are stable.

In a concentrated acidic environment, monosaccharides undergo an internal dehydration with cyclization, as a result, a furfuralic derivative is formed.

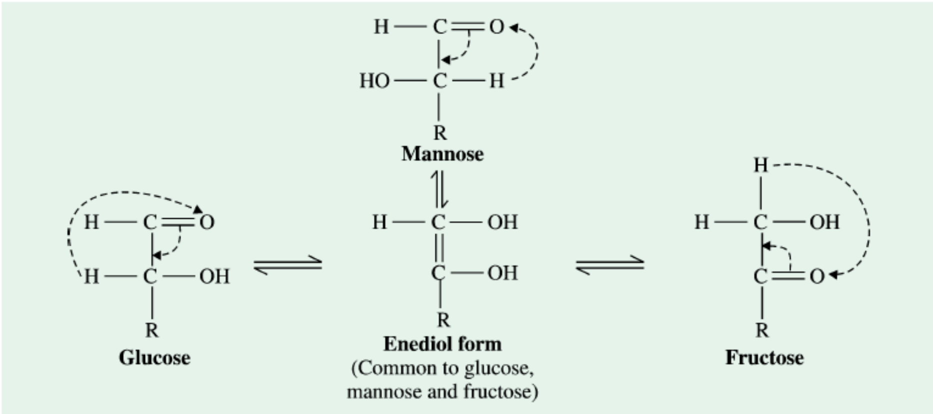


With hexoses, a 5-hydroxymethylfurfural is obtained, whereas with pentoses, a furfural product is formed.

This reaction forms the basis of some color tests for sugars, as the aldehyde products of these reactions condense with certain organic phenols and aromatic amines to give characteristic-colored products.

b. In alkaline conditions

At low temperatures : monosaccharides undergo an interconversion and epimerization on C2 in weak alkaline solutions such as $Ca(OH)_2$ and $Ba(OH)_2$.



Ex: Glucose, fructose, and mannose are interconvertible in such conditions.

The mechanism of this reaction involves enolization.

Exposed to heat: monosaccharides undergo a complete degradation.

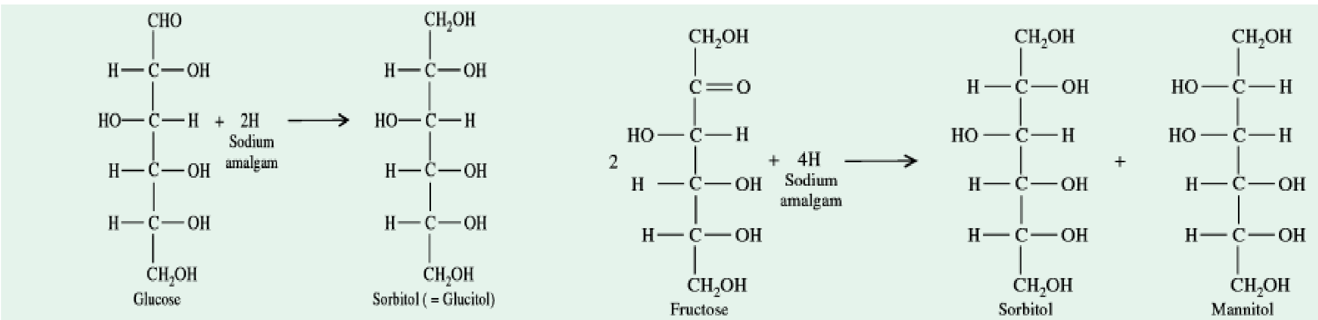
II. 2 Properties related to carbonyl group

a. Reduction

Monosaccharides' carbonyl group reduction yields their corresponding alcohols. Two pathways are possible:

- Enzymatic pathway (reversible) : by specific reductases.
- Chemical method (irreversible) : by treating with reducing agents like Na-amalgam or sodium borohydride ($NaBH_4$).

The products obtained are
- The polyalcohol corresponding to the starting aldose.
- Regarding ketoses, we obtain 2 polyalcohols that are epimeric at C2.



Examples
- D-Glucose yields D-Sorbitol (glucitol).
- D-Galactose yields D-Dulcitol.
- D-Mannose yields D-Mannitol.
- Ketosugar D-Fructose yields a mixture of D-Mannitol and D-Sorbitol.

b. <u>Oxidation</u>

➢ ***Mild oxidation: Monosaccharide* -> aldonic acid**

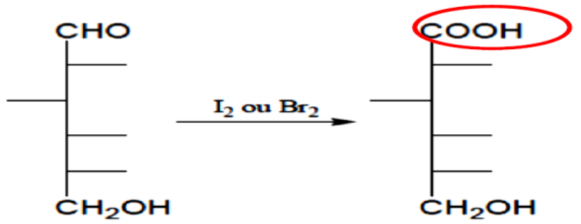
Unlike ketoses, aldoses are oxidized in mild conditions:

- Enzymatically: with specific oxidases
- Non-enzymatically : by bromine Br_2 or iodine I_2 in an alkaline medium, and very diluted nitric acid.

In such conditions, only the aldehyde group of aldoses is oxidized to produce monocarboxylic acids (CHO -> COOH). Hence, an aldonic acid is formed.

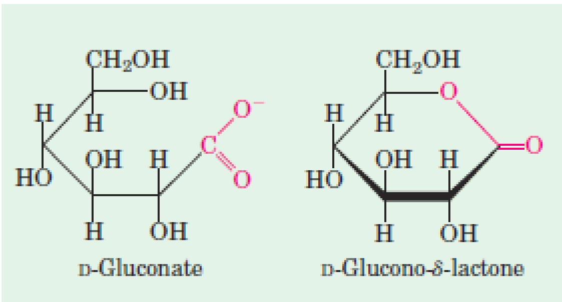
Example:

- D-glucose yields gluconic acid.
- D-mannose yields mannonic acid.
- D-galactose yields galactonic acid.



Gluconic acid

*Because aldoses can be oxidized, they are considered reducing agents. When the aldose in question is in ring form, oxidation yields a lactone instead—a cyclic ester with a carbonyl group persisting on the anomeric carbon.



**** Fehling's reagent:***

In Fehling's reagent, the cupric ion Cu^{2+} ($Cu(OH)_2$), colored in blue, readily oxidizes monosaccharides. As a result, an aldonic acid is produced with a red precipitate of Cu_2O.

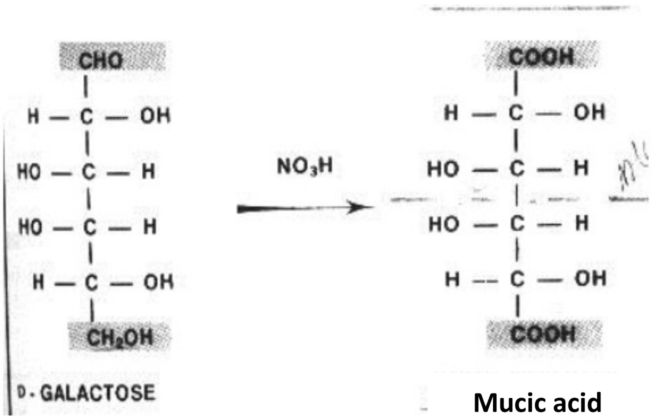
$$\underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fehling's solution}}{2\,Cu(OH)_2} \longrightarrow \underset{\text{Gluconic acid}}{C_6H_{12}O_7} + \underset{\text{Cuprous oxide}}{Cu_2O} + 2H_2O$$

This solution is widely used as an oxidizing agent preferably in quantitative sugar determinations.

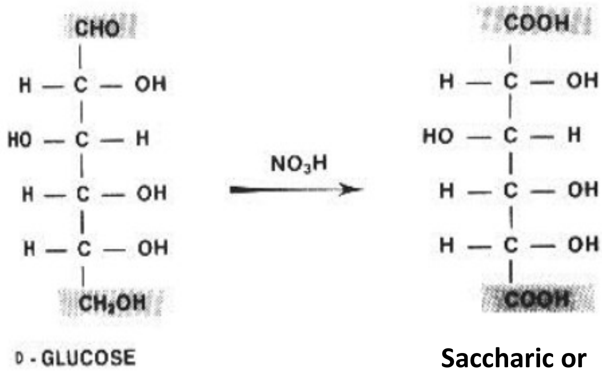
➢ ***Vigorious oxidation* : monosaccharide- > aldaric acid**

Under strong oxidation conditions (conc. HNO3+ heat) both aldehyde and primary alcohol groups are simultaneously oxidized to -COOH, Hence, it forms a dicarboxylic acid (dibasic sugar), known as saccharic acid or aldaric acid.

Example: Glucose is thus oxidized to glucosaccharic acid, mannose to mannaric acid and galactose to mucic acid.



Mucic acid



Saccharic or glucaric acid

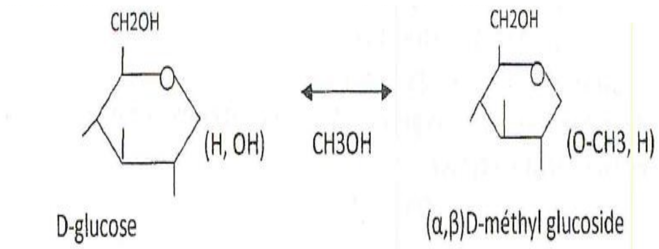
c. Addition and substitution

- **Glycoside formation**

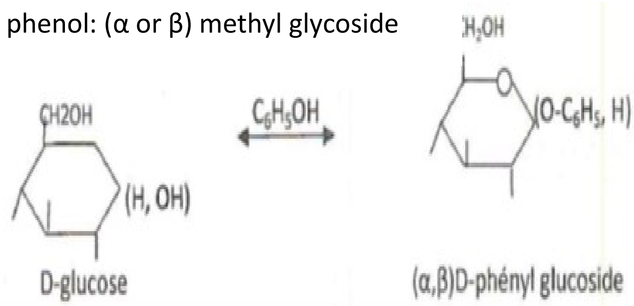
Compounds in which carbohydrate residue is attached by an acetal linkage at the anomeric carbon atom to an aglycone residue, which is a non-carbohydrate. This new linkage implies the hemiacetal function.

- **To alcohols, phenols, or sterols:** the aglycone attaches through its -OH group, forming an O-glycoside. Ex: methanol, glycolipids and glycoproteins.

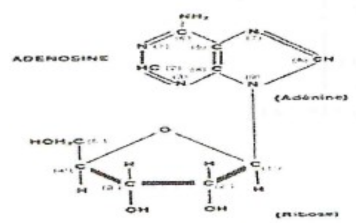
With alcohol: such as methanol = > (α or β methyl glycoside.



With phenol: (α or β) methyl glycoside



- **To ammonium and amines:** the aglycone attaches through its $-NH_2$ group, forming an N-glycoside. Ex: purine, pyrimidine (sugar nucleosides)

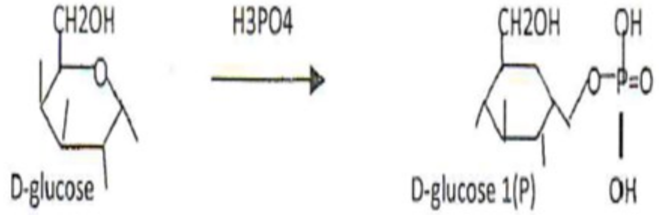


The glucosides, or glycosides in general, do not exhibit mutarotation, as the aldehyde group in them is converted to the acetal group.

Physiologically important glycosides.
· Cardiac glycosides
· Streptomycin
. Ouabain
· Glucovanillin

- **Reaction with orthophosphoric acid H_3PO_4:**

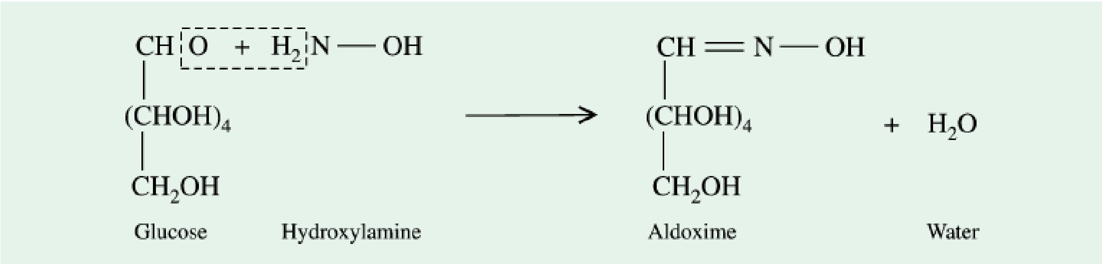
Formation of osylphosphoric acid, ex: α D-glucosyl acid 1-phosphate. Glycogenolysis product, after action of phosphorylase.



- **Reaction with hydrogen cyanide (Kiliani synthesis).**

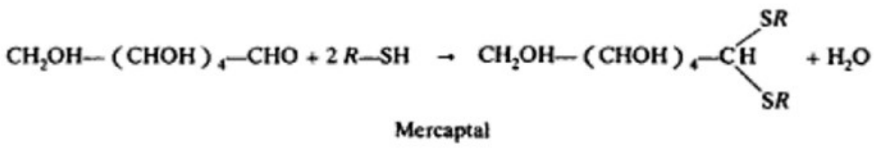
The addition of HCN to mutarotating sugars at the anomeric carbon creates a new asymmetric carbon atom and produces at the end two new sugars containing one more carbon atom than the parent sugar.

- **Reaction with hydroxylamine:** Simple sugars react with hydroxylamine to yield oximes.



- **Reaction with thiols**

In the acidic environment, the aldehydic group of aldoses combines with thiols (R-SH) and gives mercaptals, which yields the heterosides sulfur or S-heterosides.



II. 3 Properties related to alcohol group

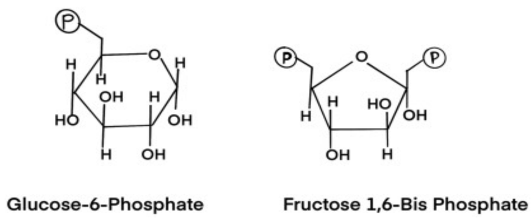
Due to the presence of many alcohol functions, monosaccharides can be converted into various alcohol derivatives, ethers, and esters.

a. Esterification

Because carbohydrates have hydroxyl groups (-OH), they are able to participate in reactions with carboxylic acids (-COOH) and carboxylic acid derivatives to form esters.

Exp1 : In the body, monosaccharides are usually present in the form of phosphate ester.

- Monophosphoric esters : glucose 6 phosphate; Cyclics: AMPc
- Diphosphoric esters: fructose 1-6 diphosphate
- Polyphosphoric esters: ATP.



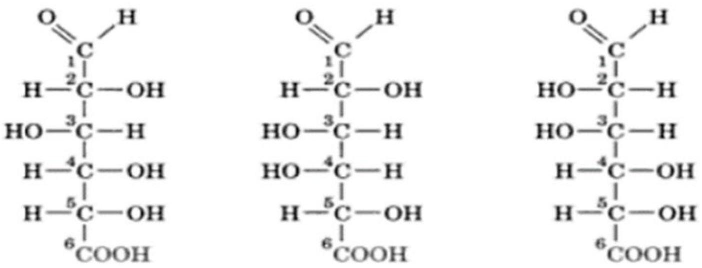
Phosphorylation of glucose is an extremely important metabolic reaction of glycolysis in which a phosphate group is transferred from ATP to glucose, thus phosphorylating glucose while forming ADP. Kinases are the enzymes that catalysis this type of reactions.

Exp2 : Sulfuric esters, found in glycosaminoglycans of connective tissue.

b. Oxidation (Monosaccharide - > Uronic acids)

When an aldose is oxidised in such a way that the primary alcohol group is converted to a –COOH group, without oxidation of the aldehyde group (protected by a methyl), a uronic acid is formed.

The name of the uronic acids is obtained by the addition of the suffix -uronic to the root of the name of the corresponding aldose.



D-glucuronic acid **D-galacturonic acid** **D-mannuronic acid**

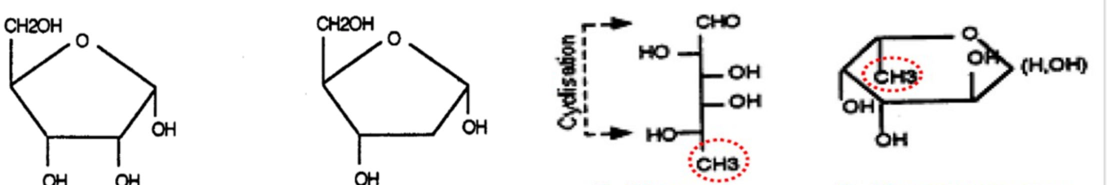
c. Reduction

Monosaccharides' hydroxyl groups can undergo a reduction to form a deoxysugar. These are sugars in which the oxygen of a -OH group has been removed, leaving the hydrogen. Thus, –CHOH or –CH2OH becomes –CH2 or –CH3.

Deoxy sugars of biological importance are:

• 2-deoxy-D-Ribose is found in nucleic acid (DNA).

• Fucose: 6-deoxy-L-Galactose is found as a constituent of glycoproteins, blood group substances and bacterial polysaccharides.



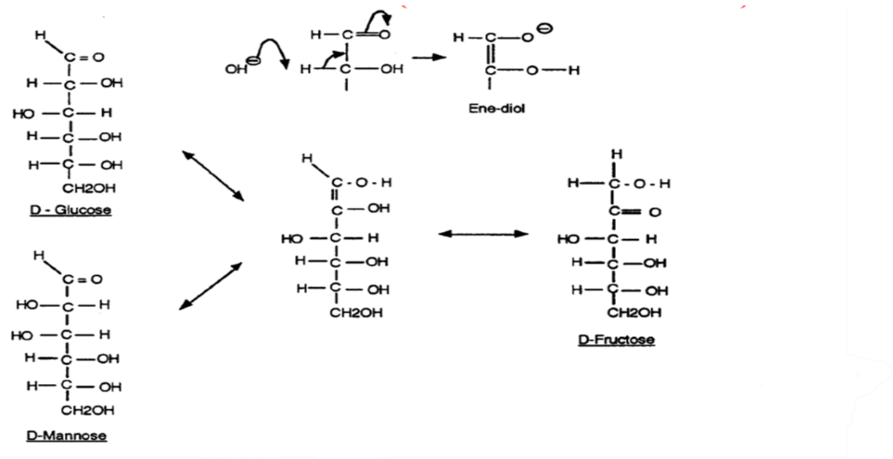
alpha - D - Ribofuranose alpha - D - 2 Desoxy ribofuranose **L-Fucose** **L-Fucopyranose**

II. 4 Properties related to the nearness between alcohol and carbonyl group

1) Interconversion

In dilute alkali, a rearrangement will occur which produce an equilibrated mixture of aldose and its epimer on C2 and its corresponding ketose. The mechanism of this reaction involves enolization

Ex: the same mixture (of glucose, fructose and mannose) is obtained even if, instead of D(+)-glucose, the starting material is D(–)-fructose or D(+)-mannose.



2) Reaction with phenyl hydrazine.

Reaction with phenyl hydrazine involves only 2 carbon atoms, the carbonyl (i.e., the aldehyde or ketone group) and the adjacent one. This reaction involves the followings step:

- One mole of phenyl hydrazine reacts with one mole of aldose (or ketose) to form a mole of hydrazone.
- With a second mole of phenyl hydrazine, the hydrazone is oxidized to aldohydrazone
- Finally, a third mole of phenyl hydrazine reacts with the aldohydrazone to produce osazone.

The osazones of reducing sugars are crystalline, yellow and usually insoluble compounds.

These have characteristic crystal structure and melting points and are, therefore, frequently used in the identification of sugars.

It may, however, be noted that glucose, fructose, and mannose would yield the same osazone owing to their similar structure in the unaffected part of the molecule (from C3 to C6).